Machine Learning Diploma

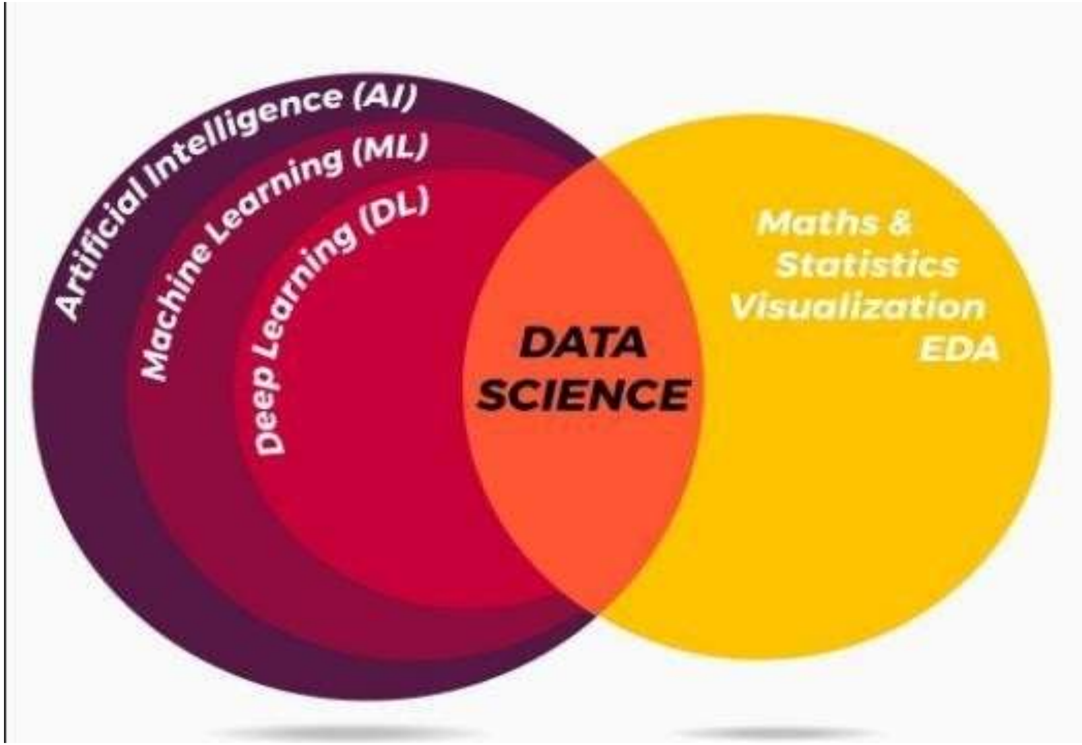
Deep Learning 1 : Artificial Neural Network

Agenda

- Difference between Deep Learning and Machine Learning
- why deep learning
- Deep Learning Application
- Perceptron Algorithm
- Artificial Neural Networks
- Deep Neural Networks

Difference between Deep Learning and Machine Learning

Deep Learning VS Machine Learning



Deep Learning VS Machine Learning

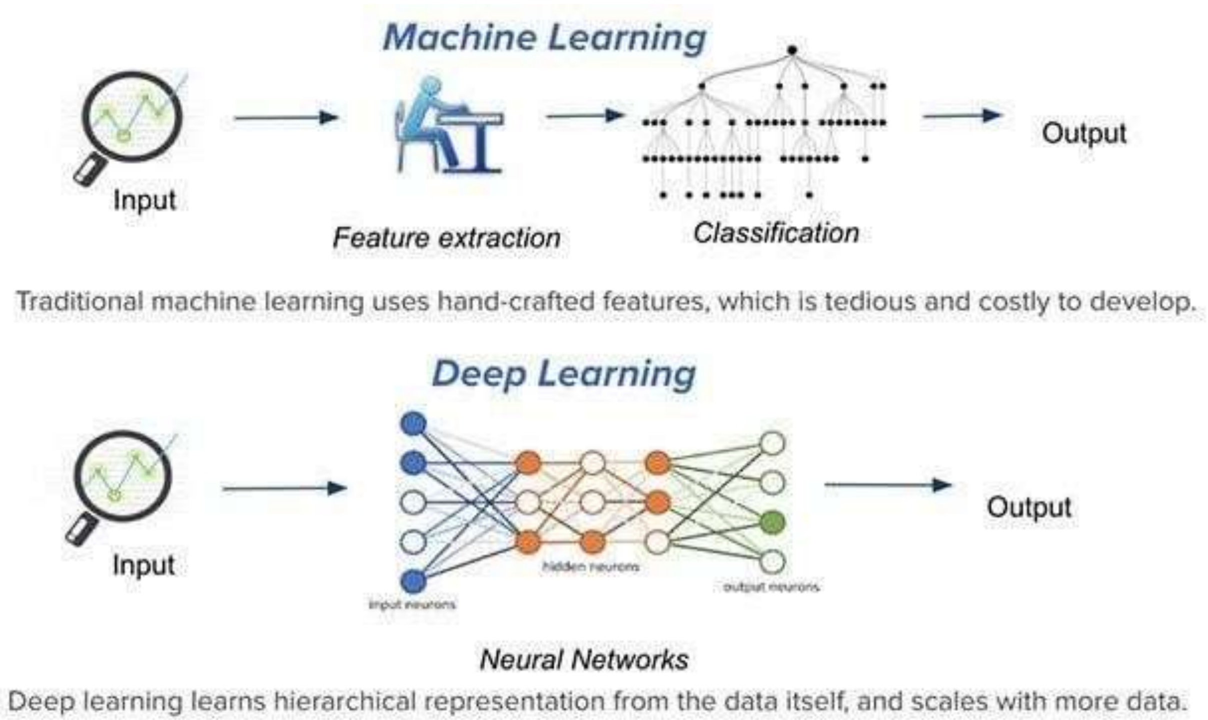
The major difference between deep learning and machine learning is its execution:

- Deep learning algorithms need a large amount of data, this is why, when the data is small deep learning algorithms don’t perform that well unlike machine learning.
- Deep learning algorithms heavily depend on high-end machines and GPUs, compare to machine learning algorithms
- Deep learning algorithms try to take high-level features from data. This is an extremely distinctive part of Deep Learning and a major step a head of Machine learning. Therefore, deep learning reduces the task of developing new feature extractor for each issue.

Deep Learning VS Machine Learning

- A deep learning algorithm takes a long time to train. This is because there are so many parameters in a deep learning algorithm than machine learning.
- Machine learning works only with sets of structured data, while deep learning works with both structured and unstructured data. Unstructured data as images, videos, texts and speech.

Deep Learning VS Machine Learning



Deep Learning VS Machine Learning

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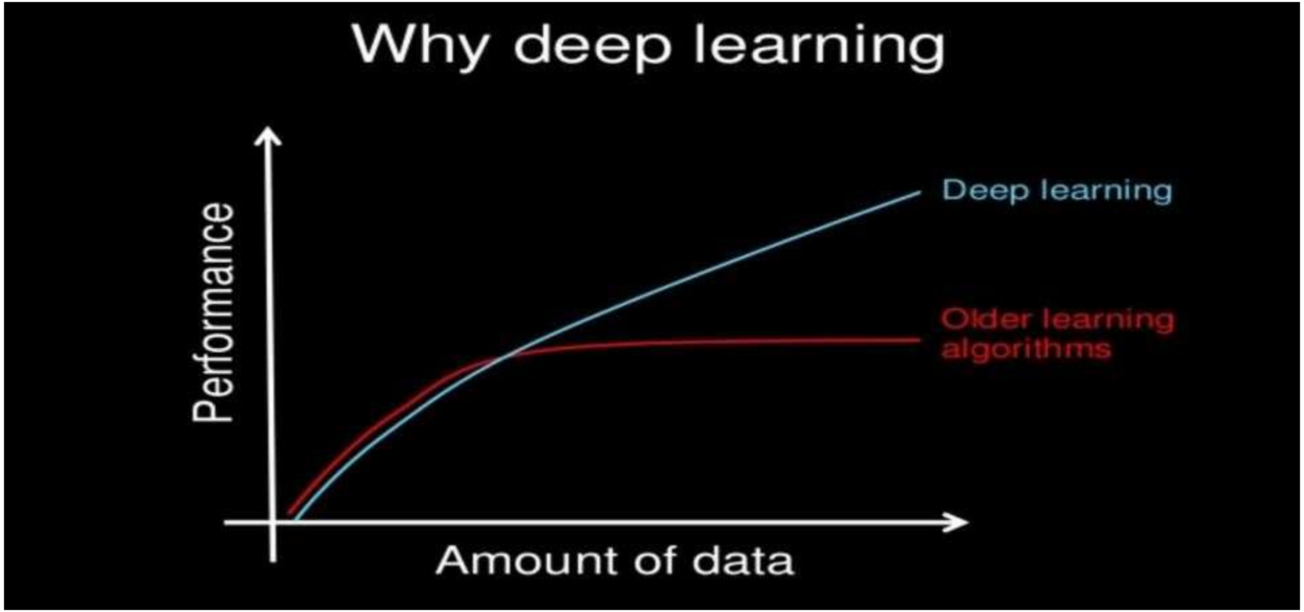
Why Deep Learning ?

- Machine learning plateaus in performance at a certain point. Even if we add more data to it, the algorithm would be not able to find more correlations to enhance its performance and results. On the other hand, deep learning which constructs artificial neural networks is getting better more and more as we construct larger neural networks and train them with more and more data.
- Thanks to fast computers and big data era we are able to have deep learning field.

Why Deep Learning ?



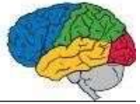
Why Deep Learning?

Important Property of Neural Networks

Results get better with

more data +
bigger models +
more computation

(Better algorithms, new insights and improved techniques always help, too!)



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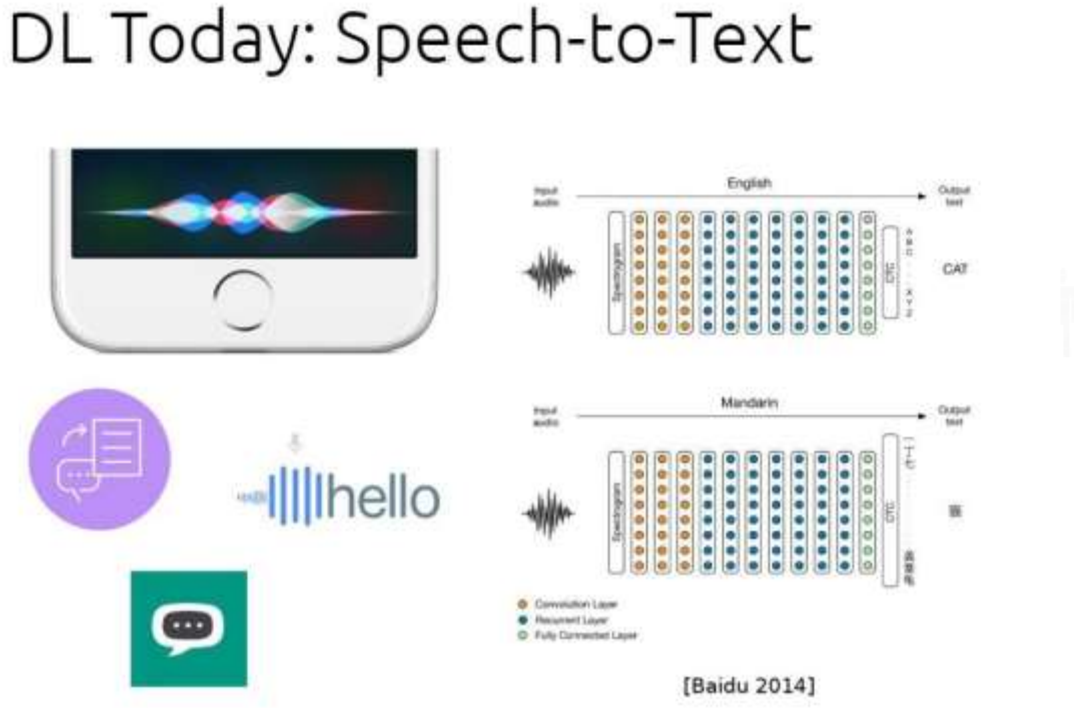
Deep Learning applications

Deep Learning Applications



Deep Learning Applications

DL Today: Vision

[Krizhevsky 2012]

[Ciresan et al. 2013]

[Faster R-CNN - Ren 2015]

[NVIDIA dev blog]

Deep Learning Applications

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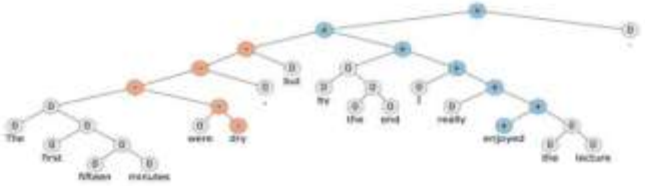
Deep Learning Applications

DL Today: NLP

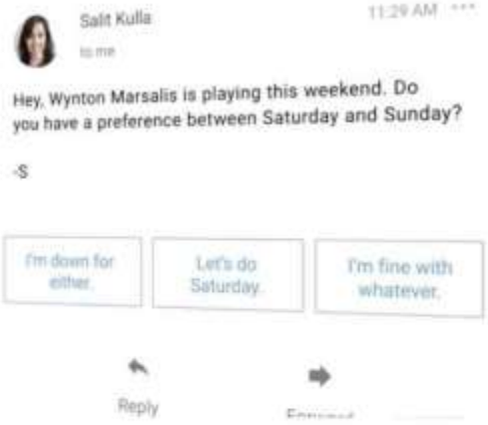
[Google Translate System - 2016]



[Socher 2015]

Deep Learning Applications

DL Today: NLP

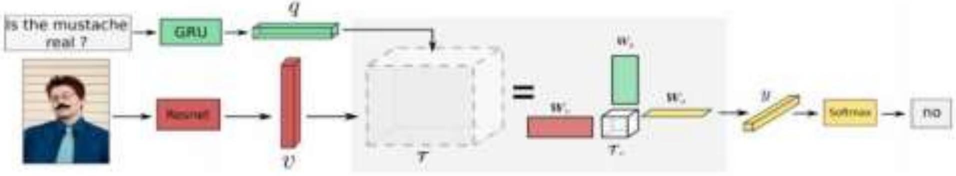


[Google Inbox Smart Reply]

[Amazon Echo / Alexa]

Deep Learning Applications

DL Today: Vision + NLP



[VQA - Mutan 2017]

"man in black shirt is playing guitar."

"construction worker in orange safety vest is working on road."

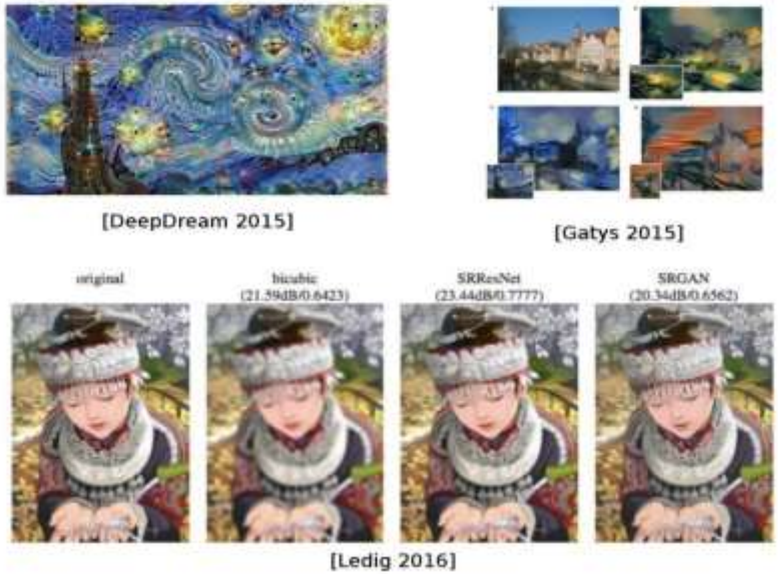
"two young girls are playing with lego toy."

"boy is doing backflip on wakeboard."

[Karpathy 2015]

Deep Learning Applications

DL Today: Image translation



Deep Learning Applications

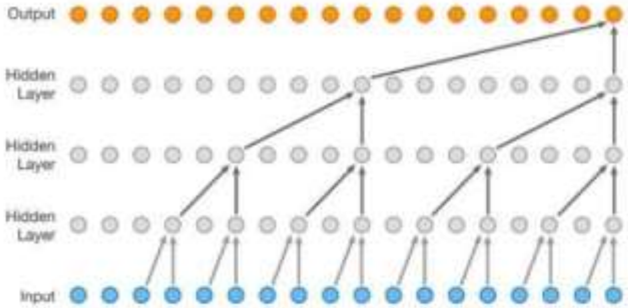
DL Today: Generative models

Sampled celebrities [Nvidia 2017]

StackGAN v2 [Zhang 2017]

Deep Learning Applications

DL Today: Generative models

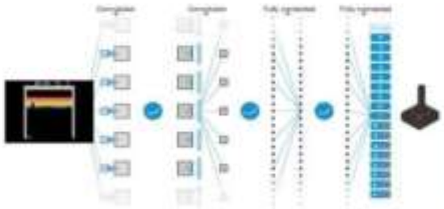


Sound generation with WaveNet [DeepMind 2017]

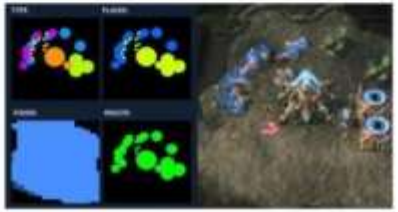
Deep Learning Applications

DL for AI in games

[Deepmind AlphaGo / Zero 2017]



[Atari Games - DeepMind 2016]



[Starcraft 2 for AI research]

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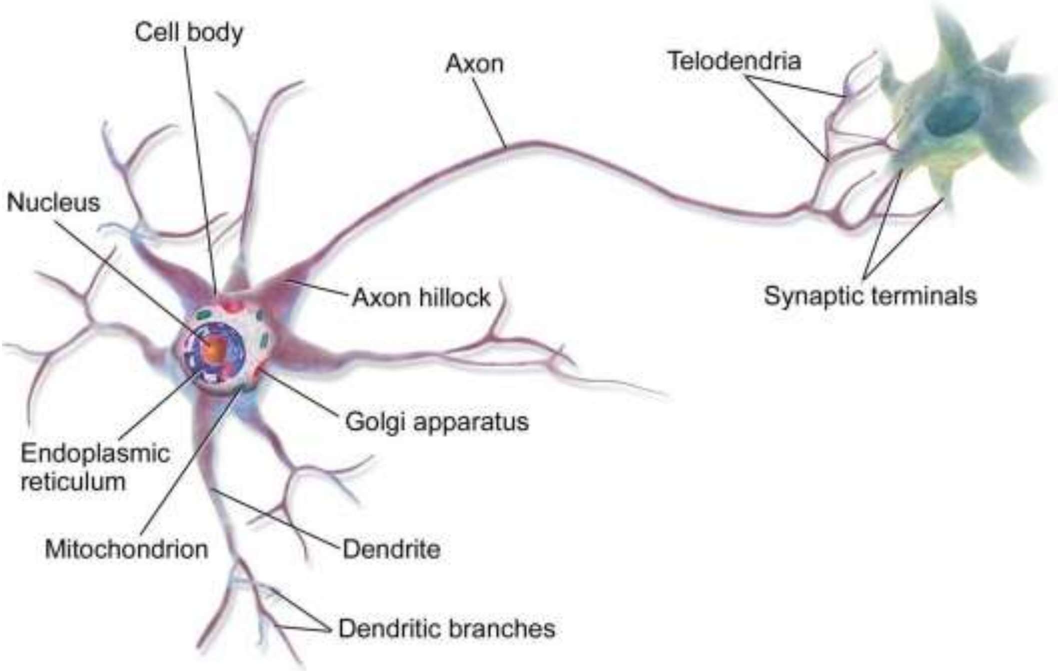
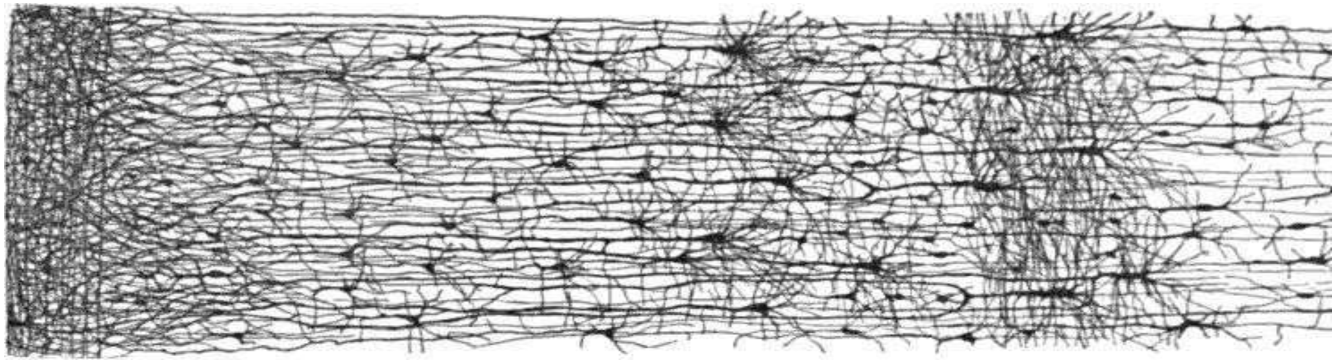
Perceptron algorithm

Biological Neuron

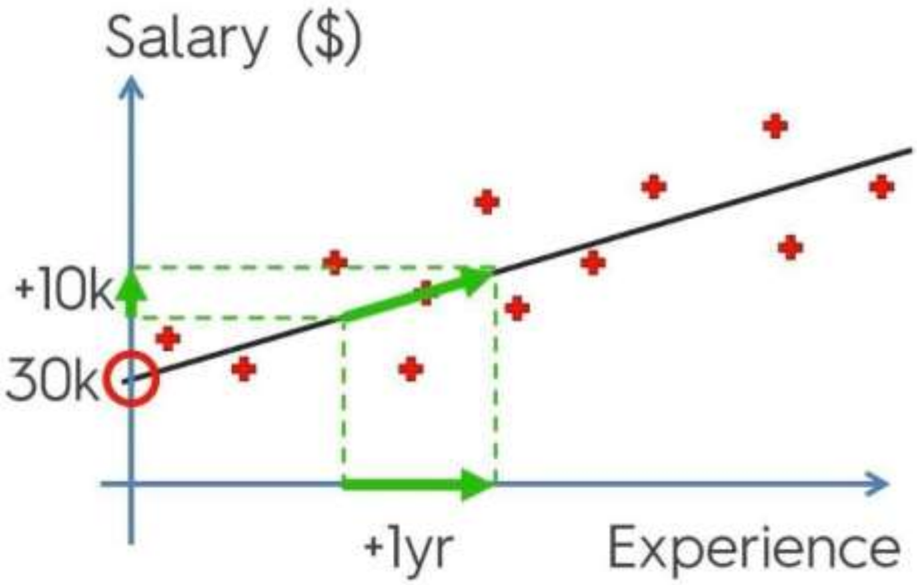


Biological Neural Network

The architecture of biological neural networks (BNNs) is still the subject of active research, but some parts of the brain have been mapped, and it seems that neurons are often organized in consecutive layers, especially in the cerebral cortex (i.e., the outer layer of your brain), as shown in the following figure.



Perceptron - Linear Regression Reminder



$y = b_0 + b_1 * x$

$\text{Salary} = b_0 + b_1 * \text{Experience}$

Perceptron - Linear Regression Reminder

Simple Linear Regression

$$y = b_0 + b_1{*}x_1$$

Multiple Linear Regression

Dependent variable (DV)

Independent variables (IVs)

$$y = b_0 + b_1{*}x_1 + b_2{*}x_2 + \dots + b_n{*}x_n$$

Constant

Coefficients

Perceptron - Linear Regression Reminder

The Logistic Function

$$y = \frac{1}{1 + e^{-f(x_1, x_2, \ldots, x_n)}} \in (0, 1)$$

where

$$f(x_1, x_2, \ldots, x_n) = a_0 + a_1 x_1 + \ldots + a_n x_n \in (-\infty, +\infty)$$

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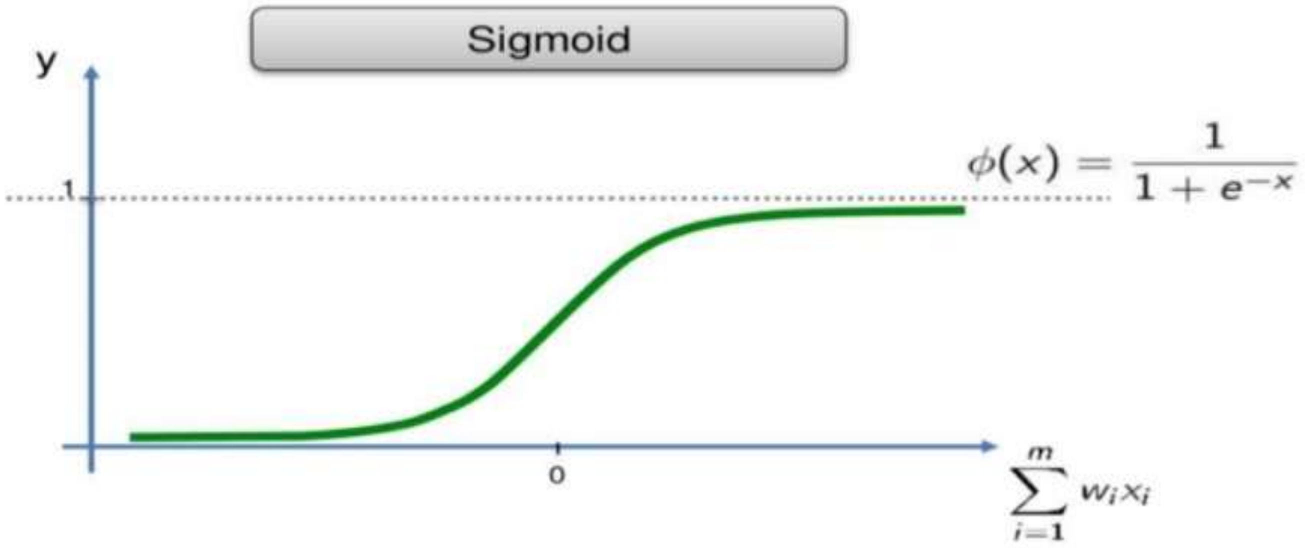
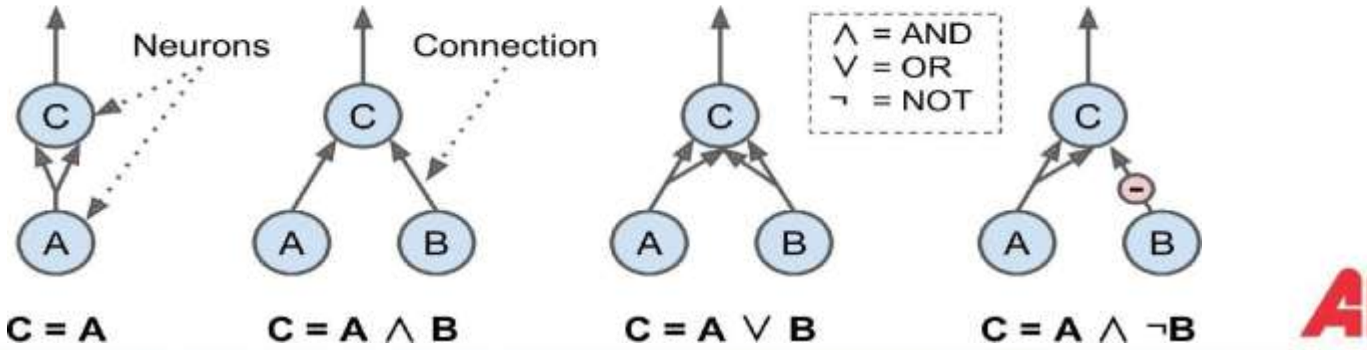
Perceptron - Linear Regression Reminder



Artificial Neuron

A proposed model of the biological neuron is made called artificial neuron where it has one or more binary inputs and one binary output. The artificial neuron activates its output when certain number of inputs is activated which was proposed in their paper, “A Logical Calculus of Ideas Immanent in Nervous Activity”, in 1943. This artificial neuron was connected with others in a network performing some logical operations.

For example, assuming that a neuron is activated when at least two of its inputs are active:



Perceptron

The step function (generally called activation function) used in perceptron is either Heaviside step function or sign function which state the following:

$$\text{heaviside}(z) = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases} \qquad \text{sgn}(z) = \begin{cases} -1 & \text{if } z < 0 \\ 0 & \text{if } z = 0 \\ +1 & \text{if } z > 0 \end{cases}$$

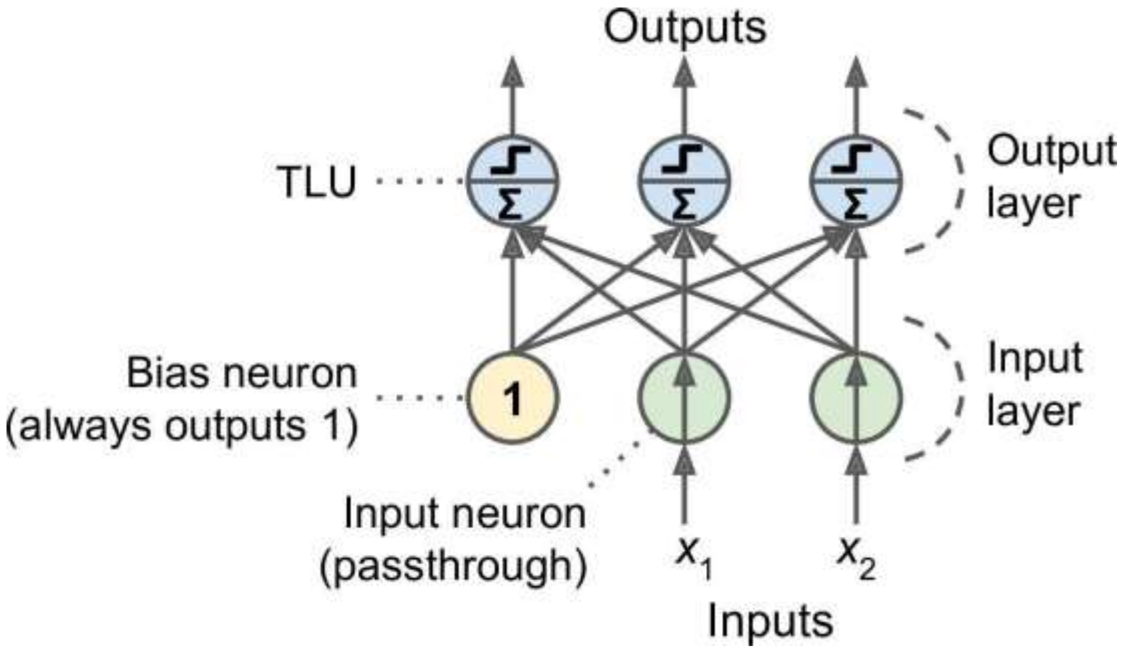
This reminds us of sigmoid function with thresholding but here is totally linear. It computes the linear combination between inputs and then we apply a type of thresholding to get our output.

Note: Similarly to machine learning part, we add a bias part where $x_0 = 1$.

Perceptron

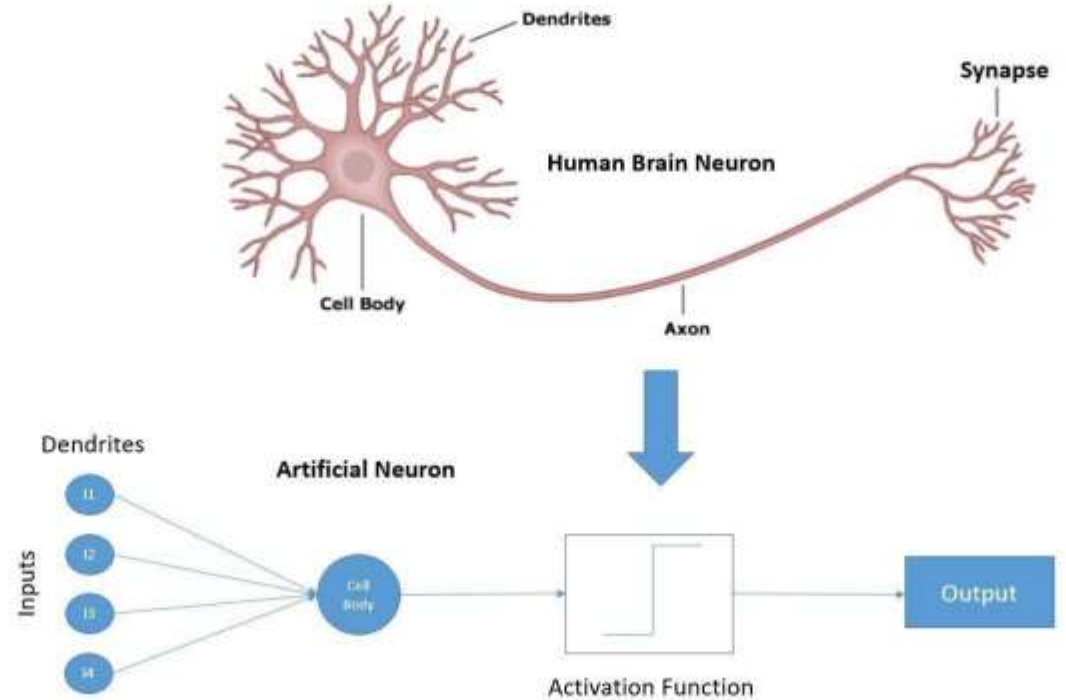
- Hence we can simply train this perceptron to classify based on some input features similarly to what we have done so far in logistic regression. Perceptron means that we have a single layer of **Threshold logic units (TLUs)** connected to inputs noting that the function taking the weighted sum is a type of step functions. When all the neurons in a layer are connected to every neuron in the previous layer (i.e., its input neurons), the layer is called a **fully connected layer, or a dense layer**.
- The inputs of the Perceptron are fed to special passthrough neurons called input neurons: they output whatever input they are fed. All the input neurons form the input layer.
- Moreover, an extra bias feature is generally added ($x_0 = 1$): it is typically represented using a special type of neuron called a bias neuron, which outputs 1 all the time. A Perceptron with two inputs and three outputs is represented in the following figure in the next slide.

Perceptron

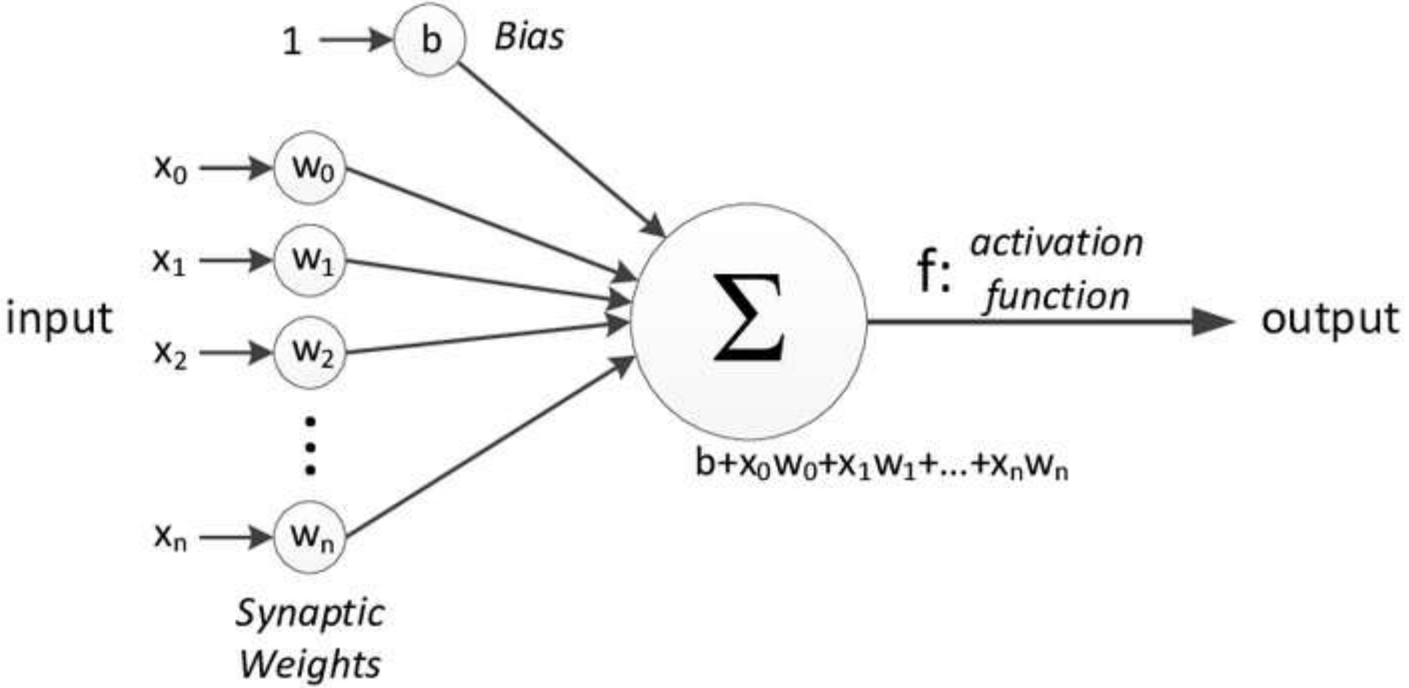


Outputs
TLU
Output layer
Bias neuron (always outputs 1)
1
Input layer
Input neuron (passthrough)
x_1
x_2
Inputs

Perceptron


Dendrites
Synapse
Human Brain Neuron
Cell Body
Axon
Dendrites
Artificial Neuron
Inputs
Cell Body
Output
Activation Function

Perceptron

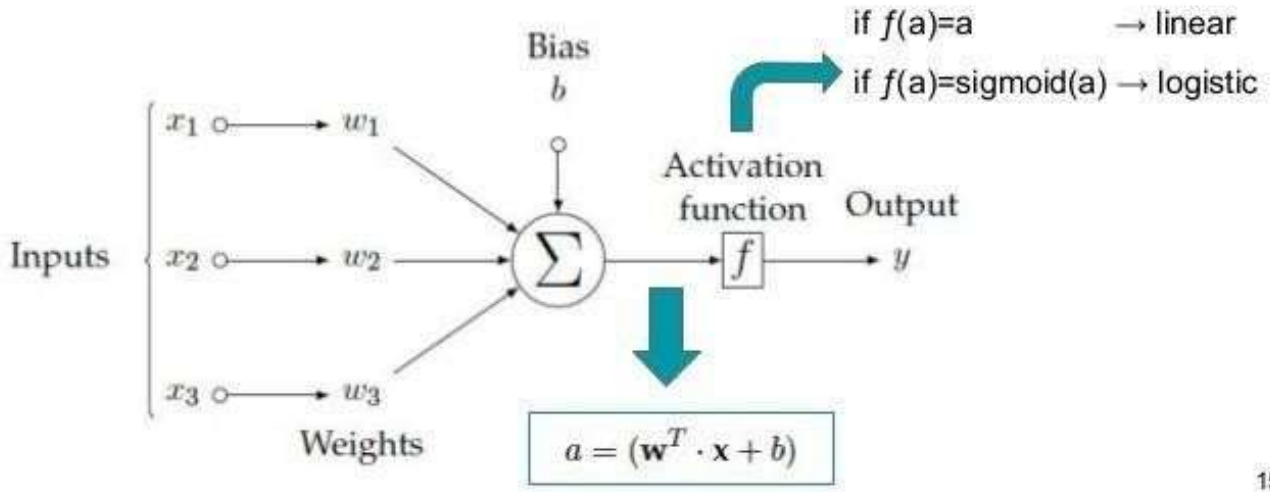


1
b
Bias
x_0
w_0
x_1
w_1
input
x_2
w_2
x_n
w_n
Synaptic Weights
Σ
$b+x_0w_0+x_1w_1+...+x_nw_n$
f: activation function
output

Perceptron

The Perceptron (Neuron)

The Perceptron can represent both linear & logistic regression:



Perceptron

Linear algebra can be used here to get the output of several instances at once.

$$h_{\mathbf{W},\mathbf{b}}(\mathbf{X}) = \phi(\mathbf{X}\mathbf{W} + \mathbf{b})$$

Where:

- X represents the matrix of the input features(row per instance, column per feature)
- W matrix is all connections from input features (except bias) to the output neurons. Row per feature and column per output neuron.
- b is a vector containing all weights between bias neuron and output neurons.
- Φ is the activation function (step function) applied to the weighted sum.

Implement Perceptron From Scratch

Artificial Neural Network

Artificial Neural Network (ANN)

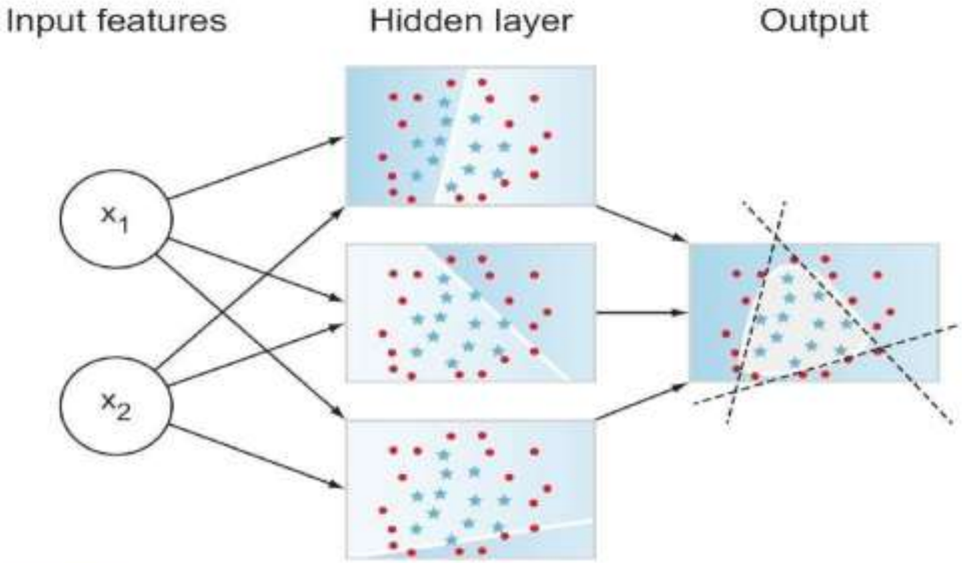
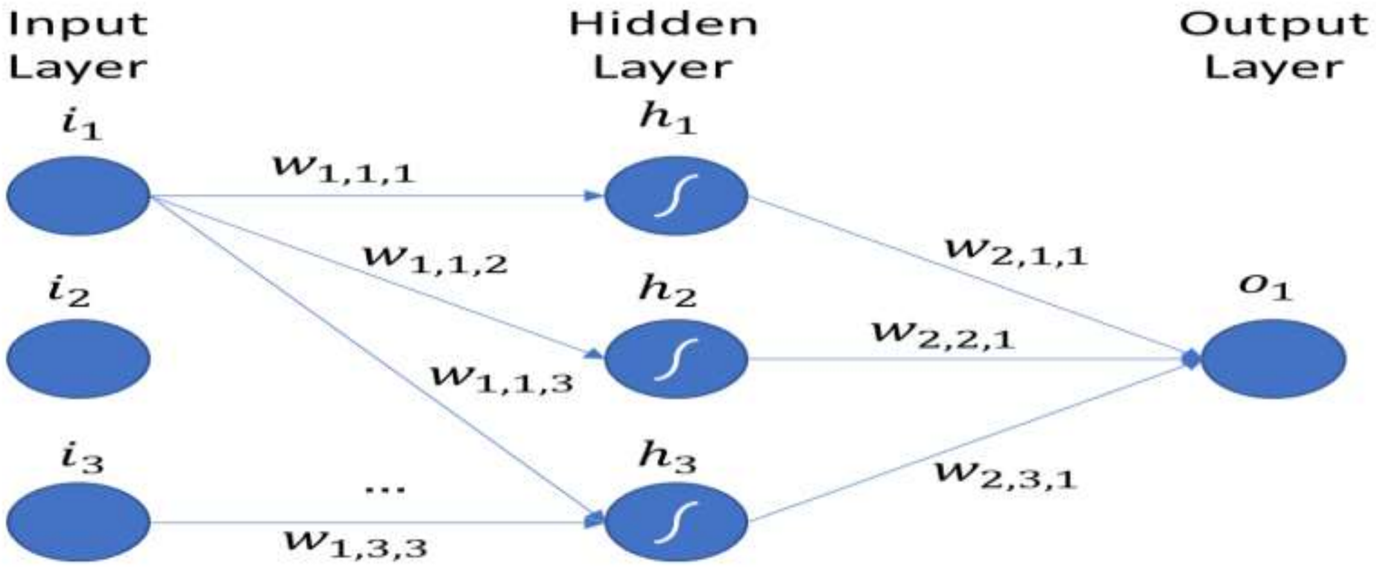


Figure 2.10

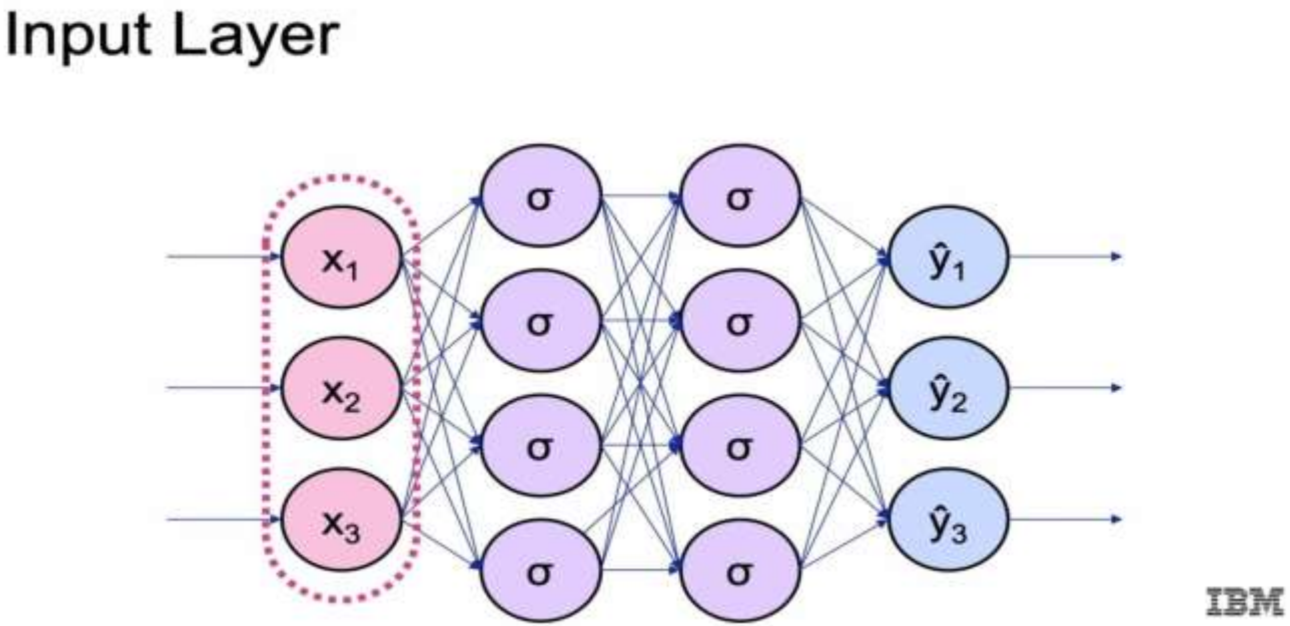
Artificial Neural Network (ANN)

- Multilayer Perceptron MLP consists of a passthrough layer of inputs similar to perceptron but instead of passing to another layer with output neurons, we pass them to one or multiple layers consisting of TLUs that we call hidden layers then passing to the final output layer. The layers closer to the input are called the lower layers (low level features layer) and as we go deeper to the output layer we call it higher layers (high level features layer).
- **Note:** Every layer in the hidden layers in addition to the input layer are
- having biases and is fully connected to the next layer.

Artificial Neural Network (ANN)



Artificial Neural Network (ANN)



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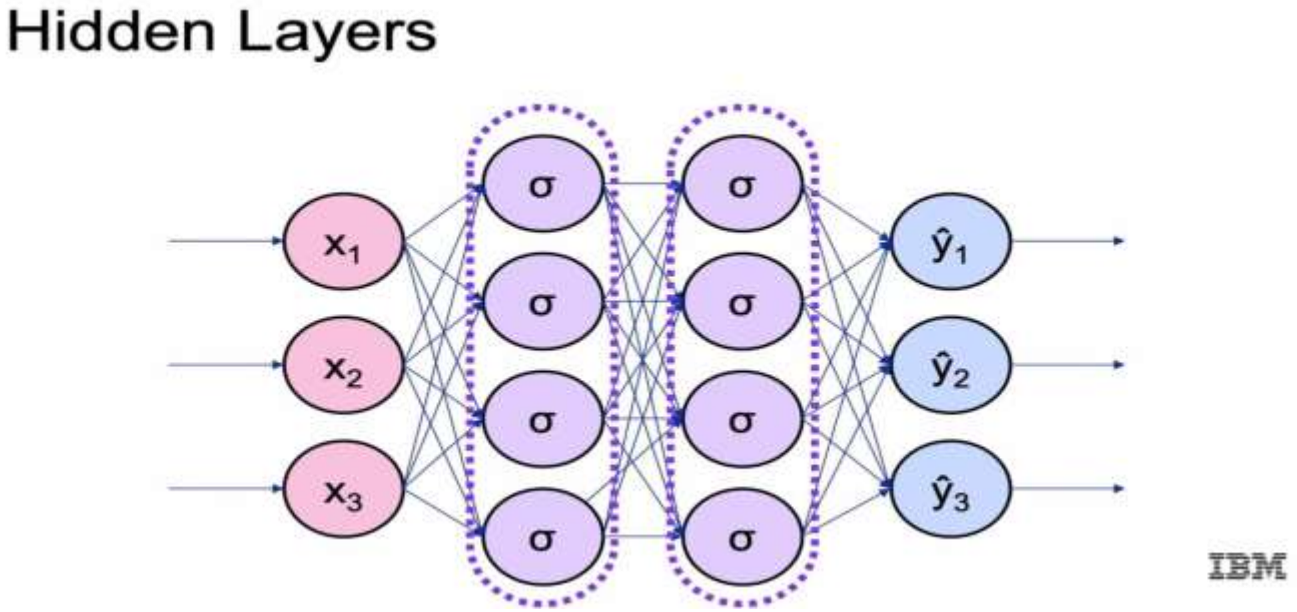
Artificial Neural Network (ANN)



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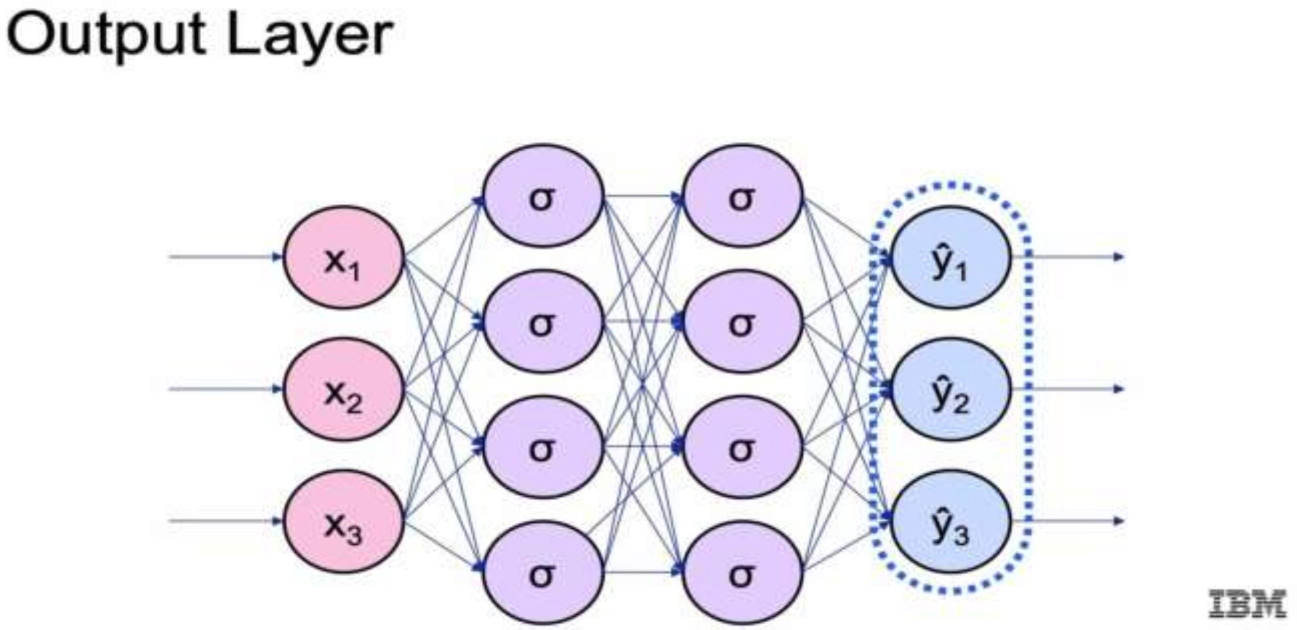
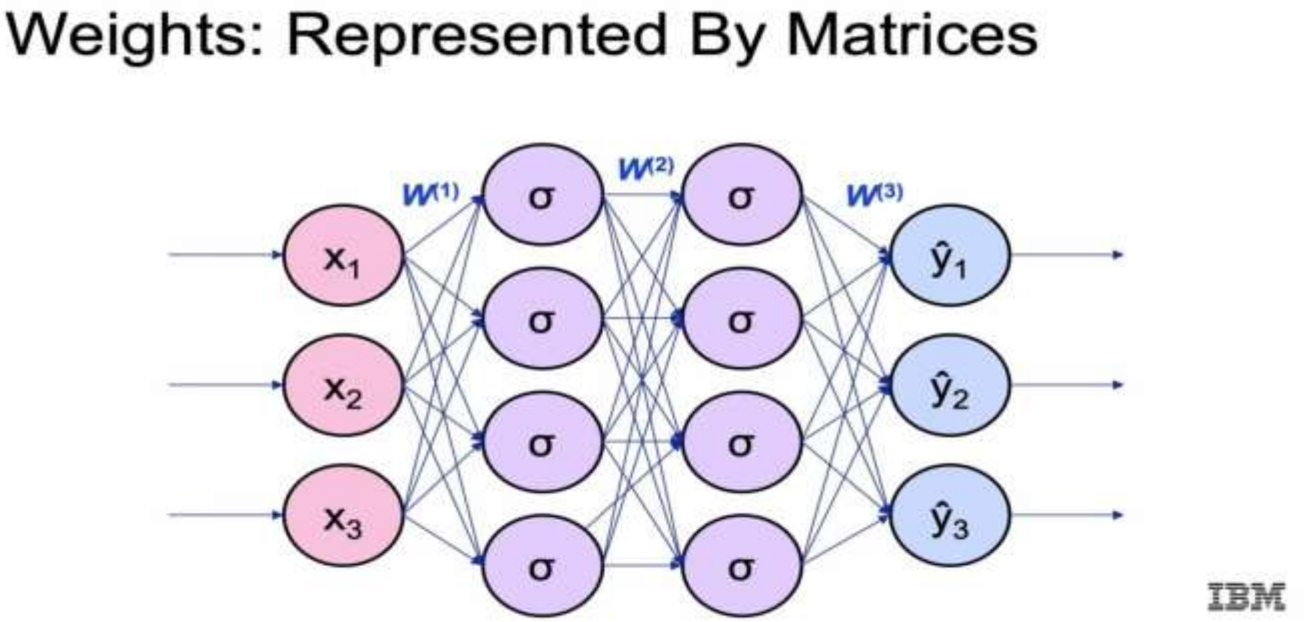
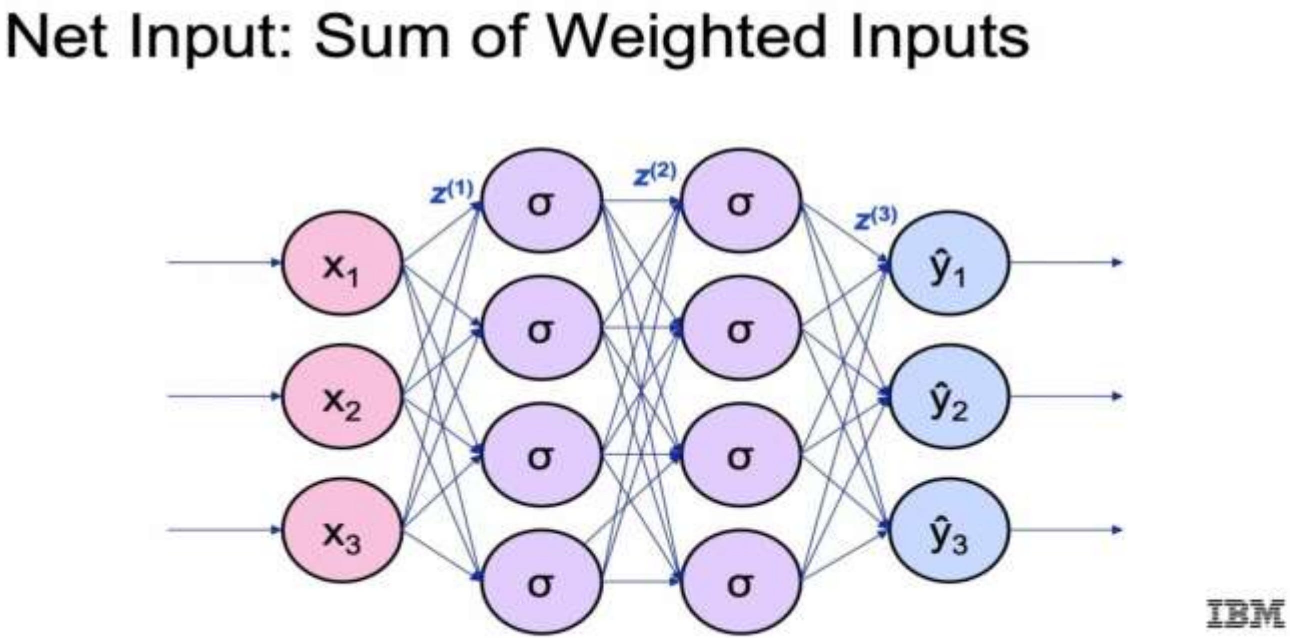
Artificial Neural Network (ANN)



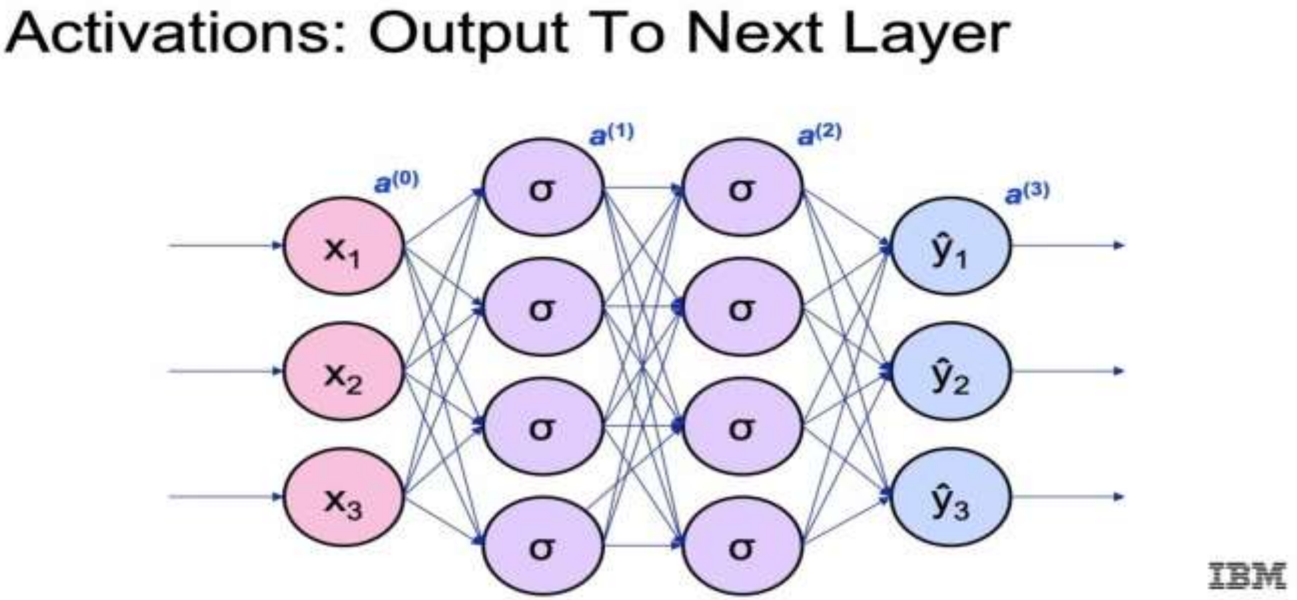
Artificial Neural Network (ANN)



Artificial Neural Network (ANN)



Artificial Neural Network (ANN)



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Artificial Neural Network (ANN)

Matrix Representation of Computation

For a single data point (instance):

$W^{(1)}$ is a 3x4 matrix

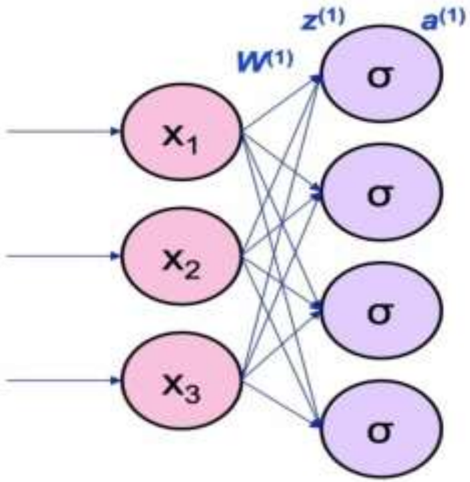
$z^{(1)}$ is a 4-vector

$a^{(1)}$ is a 4-vector

$x \quad (x = a^{(0)})$

$$z^{(1)} = xW^{(1)}$$

$$a^{(1)} = \sigma(z^{(1)})$$



Any Questions?

THANK YOU!

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